at all levels, from government agencies to online education providers, traditional universities, and student enrollees. Such a large problem is typically composed of many smaller problems, some of which can benefit from a blockchain solution. The blockchain-based solution you'll explore in this chapter works within a large, conventional system, improving the scalability and efficacy of a particular subsystem.

You may wonder what educational credentials and certificates have to do with blockchain. A lot. You are fortunate to be able to enjoy online lessons on many subjects and skills through a variety of digital media and in classrooms. Although the methods of educational delivery have diversified vastly in the past decade, most methods of evaluating these credentials for certification and degree auditing are still performed manually or by a legacy application, using the records stored in traditional student databases. What we want is an independent application. Students and other key stakeholders (advisers) should be able to verify credentials without the help of the centralized databases, such as to keep track of students' progress through a certificate or a degree program.

You could write a traditional web application that addresses this problem by using the data from a central student database. But imagine this situation:

- The participants in the program are decentralized.
- The courses or credentials originate from many educational settings (such as online courses and work experiences).
- The participants are not associated with any traditional institution, such as a university.